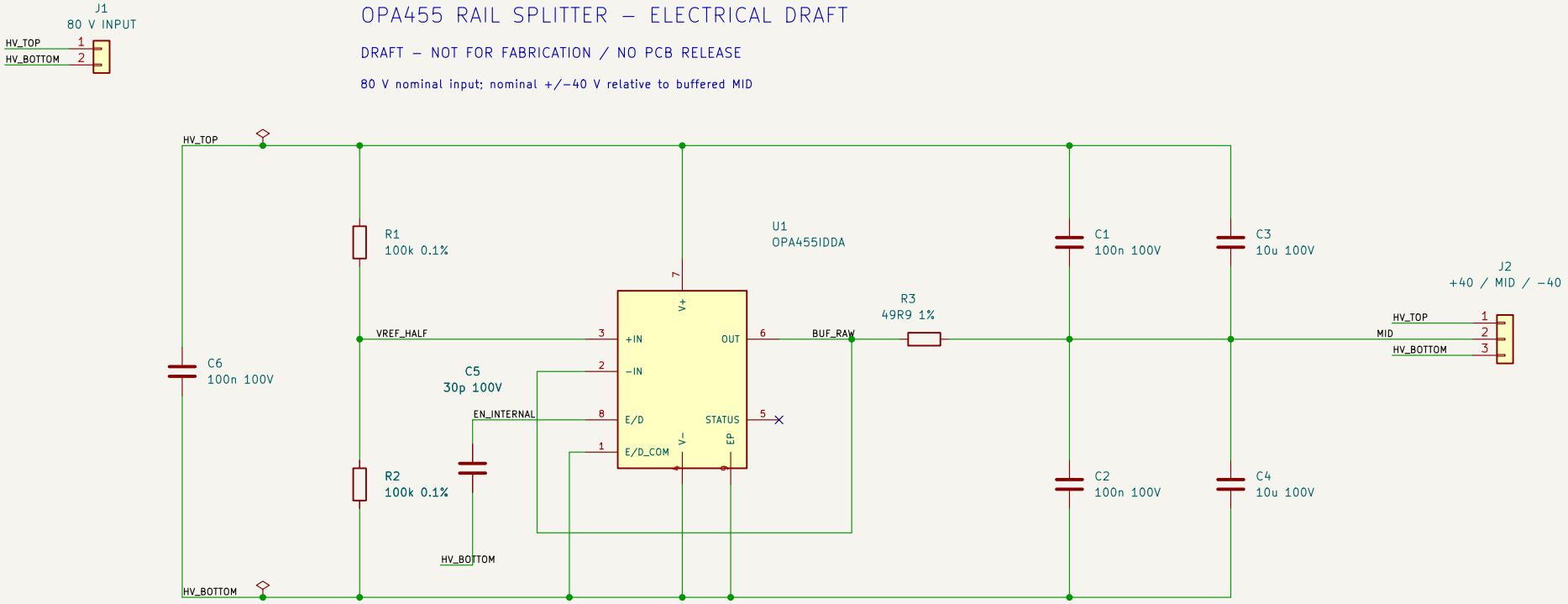
OPA455 RAIL SPLITTER — ELECTRICAL DRAFT

DRAFT — NOT FOR FABRICATION / NO PCB RELEASE

80 V nominal input; nominal +/−40 V relative to buffered MID



J1.2 = GAPTEC pin 5 = boost GND = Howland −40 V
J2.2 = signal ISO_GND = Howland GND
Do not join MID to HV_BOTTOM. Each channel stays isolated.

R3 is provisional. Feedback is BEFORE R3.
5 mA DC imbalance causes about 0.25 V midpoint shift.
C3/C4 are nominal 10u; verify bias−dependent capacitance.

C5 filters the internal E/D pullup; E/D_COM ties to HV_BOTTOM.
STATUS is intentionally unused. EP pin 9 must solder to HV_BOTTOM.
C6 must be directly beside U1 supply pins on the eventual PCB.

Required before release: loop/step−load validation, startup and shutdown, load imbalance and thermal tests, protection, exact BOM, PCB isolation/layout review. ERC success does not validate stability.

R1/R2: 16 mW each at 80 V; 32 mW total. Divider tolerance alone: +/−40 mV at MID.
Assume one channel, +/−5 mA midpoint imbalance for initial evaluation. This is not a validated load limit.

NOT FOR FABRICATION; stability and system power unvalidated

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Title: OPA455 buffered midpoint — electrical draft

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